

Numerical Methods

Week 12

Ordinary Differential Equations (ODEs)



ODE Outline

- ▣ Introduction to ODEs
- ▣ Taylor series methods
- ▣ Midpoint and Heun's method
- ▣ Runge-Kutta methods

Taylor Series Methods



Learning Objectives

- ❑ Derive Euler formula using the Taylor series expansion.
- ❑ Solve the first order ODEs using Euler method.
- ❑ Assess the error level when using Euler method.
- ❑ Appreciate different types of errors in the numerical solution of ODEs.
- ❑ Improve Euler method using higher-order Taylor Series.

Taylor Series Method

The problem to be solved is a first order ODE:

$$\frac{dy(x)}{dx} = f(x, y), \quad y(x_0) = y_0$$

Estimates of the solution at different base points:

$$y(x_0 + h), y(x_0 + 2h), y(x_0 + 3h), \dots$$

are computed using the truncated Taylor series expansions.

Taylor Series Expansion

Truncated Taylor Series Expansion

$$y(x_0 + h) \approx \sum_{k=0}^n \frac{h^k}{k!} \left(\frac{d^k y}{dx^k} \bigg|_{x=x_0, y=y_0} \right)$$
$$\approx y(x_0) + h \frac{dy}{dx} \bigg|_{x=x_0, y=y_0} + \frac{h^2}{2!} \frac{d^2 y}{dx^2} \bigg|_{x=x_0, y=y_0} + \dots + \frac{h^n}{n!} \frac{d^n y}{dx^n} \bigg|_{x=x_0, y=y_0}$$

The n^{th} order Taylor series method uses the n^{th} order Truncated Taylor series expansion.

Euler Method

- First order Taylor series method is known as Euler Method.
- Only the constant term and linear term are used in the Euler method.
- The error due to the use of the truncated Taylor series is of order $O(h^2)$.

First Order Taylor Series Method

(Euler Method)

$$y(x_0 + h) = y(x_0) + h \left. \frac{dy}{dx} \right|_{\substack{x=x_0, \\ y=y_0}} + O(h^2)$$

Notation:

$$x_n = x_0 + nh, \quad y_n = y(x_n),$$

$$\left. \frac{dy}{dx} \right|_{\substack{x=x_i, \\ y=y_i}} = f(x_i, y_i)$$

Euler Method

$$y_{i+1} = y_i + h f(x_i, y_i)$$

Euler Method

Problem:

Given the first order ODE : $\dot{y}(x) = f(x, y)$

with the initial condition: $y_0 = y(x_0)$

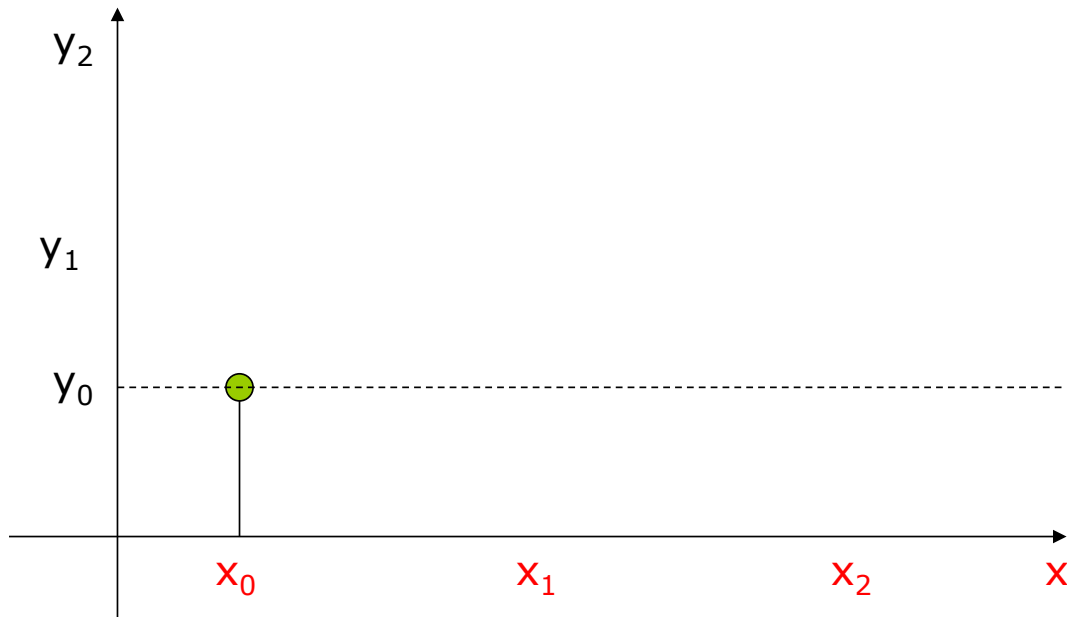
Determine: $y_i = y(x_0 + ih)$ *for* $i = 1, 2, \dots$

Euler Method:

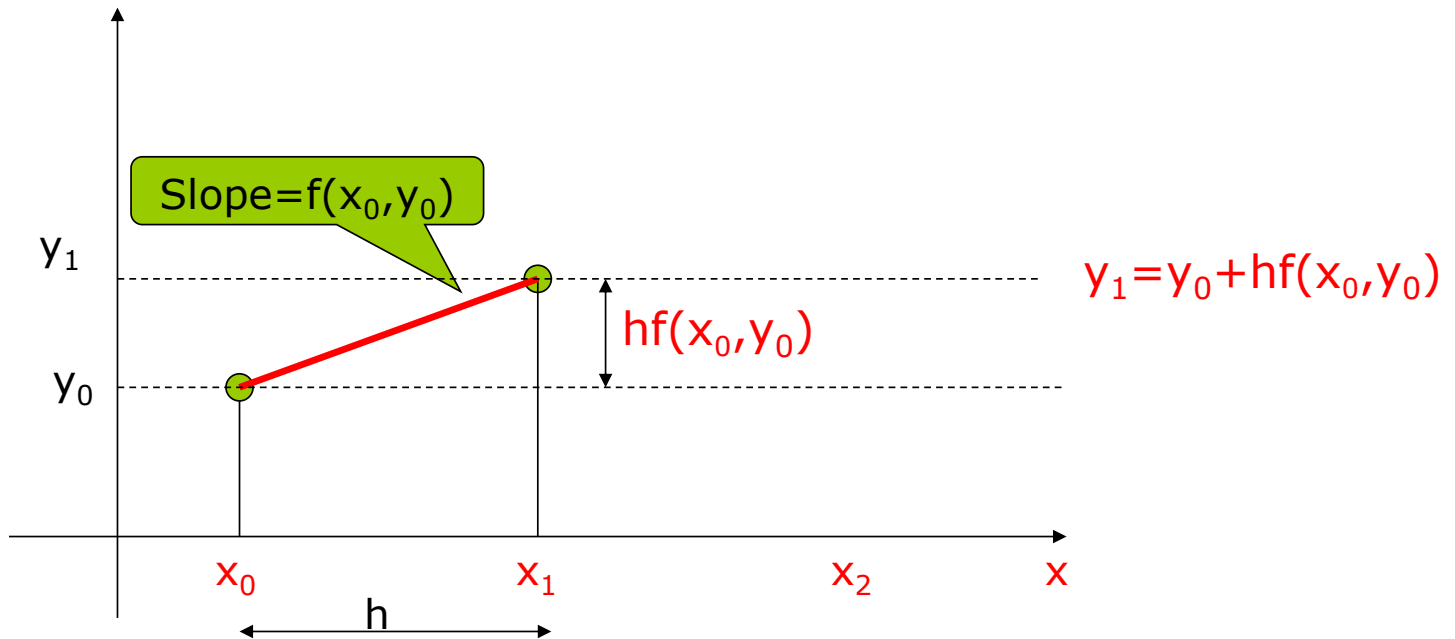
$$y_0 = y(x_0)$$

$$y_{i+1} = y_i + h f(x_i, y_i) \quad \text{for } i = 1, 2, \dots$$

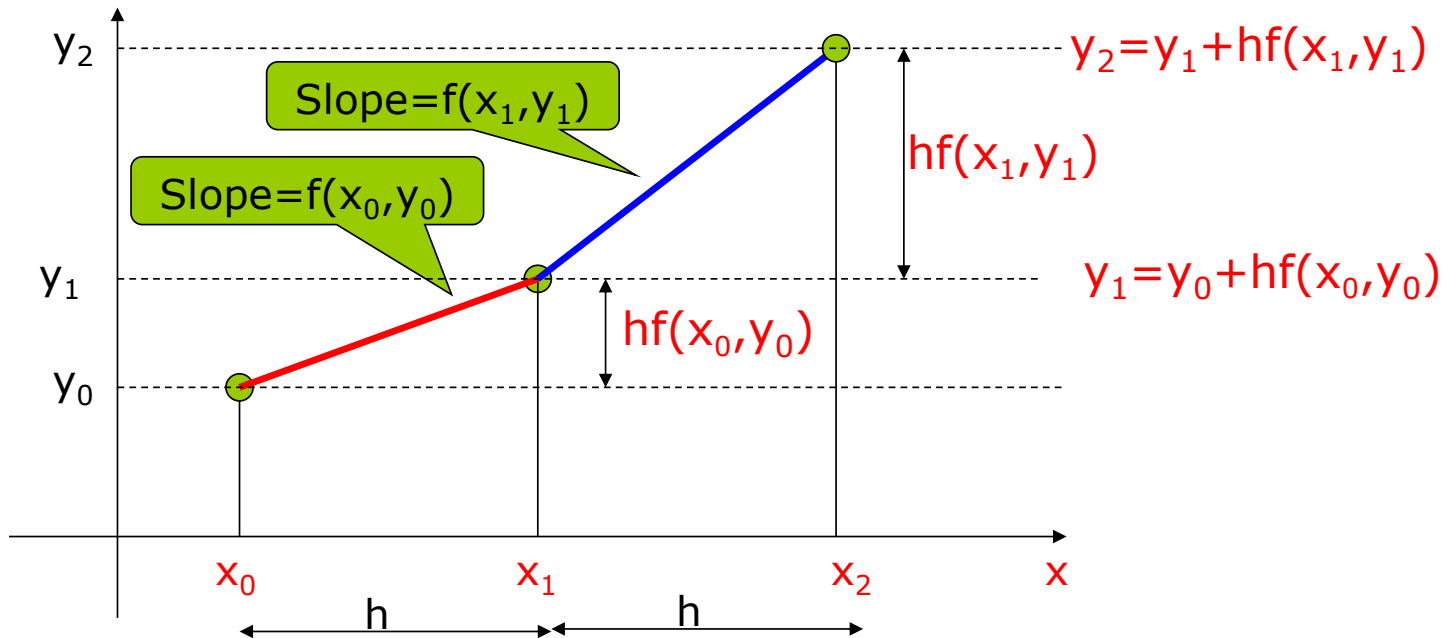
Interpretation of Euler Method



Interpretation of Euler Method



Interpretation of Euler Method



Example 1

Use Euler method to solve the ODE:

$$\frac{dy}{dx} = 1 + x^2, \quad y(1) = -4$$

to determine $y(1.01)$, $y(1.02)$ and $y(1.03)$.

Example 1

$$f(x, y) = 1 + x^2, \quad x_0 = 1, \quad y_0 = -4, \quad h = 0.01$$

Euler Method

$$y_{i+1} = y_i + h f(x_i, y_i)$$

$$\text{Step1: } y_1 = y_0 + h f(x_0, y_0) = -4 + 0.01(1 + (1)^2) = -3.98$$

$$\text{Step2: } y_2 = y_1 + h f(x_1, y_1) = -3.98 + 0.01(1 + (1.01)^2) = -3.9598$$

$$\text{Step3: } y_3 = y_2 + h f(x_2, y_2) = -3.9598 + 0.01(1 + (1.02)^2) = -3.9394$$

Example 1

$$f(x, y) = 1 + x^2, \quad x_0 = 1, \quad y_0 = -4, \quad h = 0.01$$

Summary of the result:

i	x_i	y_i
0	1.00	-4.00
1	1.01	-3.98
2	1.02	-3.9595
3	1.03	-3.9394

Example 1

$$f(x, y) = 1 + x^2, \quad x_0 = 1, \quad y_0 = -4, \quad h = 0.01$$

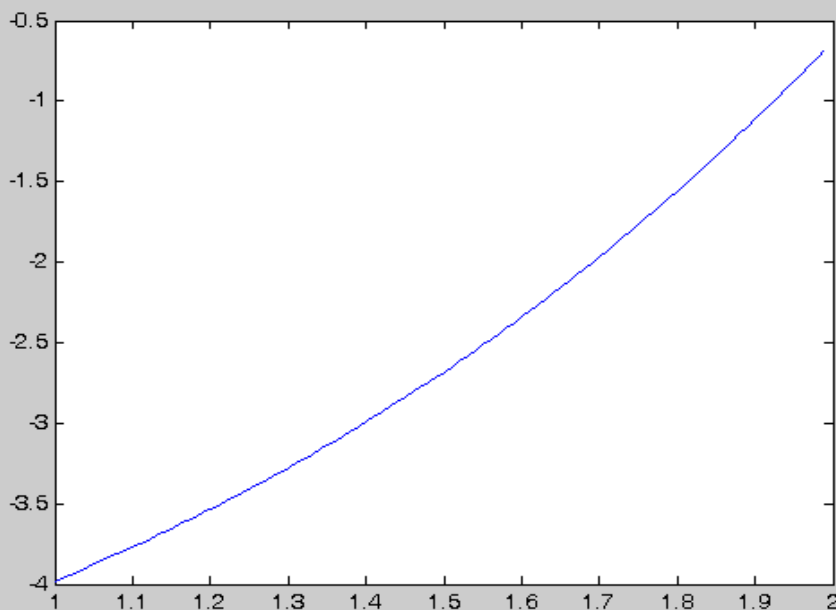
Comparison with true value:

i	x_i	y_i	True value of y_i
0	1.00	-4.00	-4.00
1	1.01	-3.98	-3.97990
2	1.02	-3.9595	-3.95959
3	1.03	-3.9394	-3.93909

Example 1

$$f(x, y) = 1 + x^2, \quad x_0 = 1, \quad y_0 = -4, \quad h = 0.01$$

A graph of the
solution of the
ODE for
 $1 < x < 2$



Types of Errors

- **Local truncation error:**

Error due to the use of truncated Taylor series to compute $x(t+h)$ in one step.

- **Global Truncation error:**

Accumulated truncation over many steps.

- **Round off error:**

Error due to finite number of bits used in representation of numbers. This error could be accumulated and magnified in succeeding steps.

Second Order Taylor Series Methods

$$\text{Given } \frac{dy(x)}{dx} = f(y, x), \quad y(x_0) = y_0$$

Second order Taylor Series method

$$y_{i+1} = y_i + h \frac{dy}{dx} + \frac{h^2}{2!} \frac{d^2 y}{dx^2} + O(h^3)$$

$\frac{d^2 y}{dx^2}$ needs to be derived analytically.

Third Order Taylor Series Methods

$$\text{Given } \frac{dy(x)}{dx} = f(y, x), \quad y(x_0) = y_0$$

Third order Taylor Series method

$$y_{i+1} = y_i + h \frac{dy}{dx} + \frac{h^2}{2!} \frac{d^2 y}{dx^2} + \frac{h^3}{3!} \frac{d^3 y}{dx^3} + O(h^4)$$

$\frac{d^2 y}{dx^2}$ and $\frac{d^3 y}{dx^3}$ need to be derived analytically.

High Order Taylor Series Methods

$$\text{Given } \frac{dy(x)}{dx} = f(y, x), \quad y(x_0) = y_0$$

n^{th} order Taylor Series method

$$y_{i+1} = y_i + h \frac{dy}{dx} + \frac{h^2}{2!} \frac{d^2 y}{dx^2} + \dots + \frac{h^n}{n!} \frac{d^n y}{dx^n} + O(h^{n+1})$$

$\frac{d^2 y}{dx^2}, \frac{d^3 y}{dx^3}, \dots, \frac{d^n y}{dx^n}$ need to be derived analytically.

Higher Order Taylor Series Methods

- ❑ High order Taylor series methods are more accurate than Euler method.
- ❑ But, the 2nd, 3rd, and higher order derivatives need to be derived analytically which may not be easy.

Example 2

Second order Taylor Series Method

Use Second order Taylor Series method to solve :

$$\frac{dx}{dt} + 2x^2 + t = 1, \quad x(0) = 1, \quad \text{use } h = 0.01$$

What is : $\frac{d^2x(t)}{dt^2}$?

Example 2

Use Second order Taylor Series method to solve :

$$\frac{dx}{dt} + 2x^2 + t = 1, \quad x(0) = 1, \quad \text{use } h = 0.01$$

$$\frac{dx}{dt} = 1 - 2x^2 - t$$

$$\frac{d^2x(t)}{dt^2} = 0 - 4x \frac{dx}{dt} - 1 = -4x(1 - 2x^2 - t) - 1$$

$$x_{i+1} = x_i + h(1 - 2x_i^2 - t_i) + \frac{h^2}{2}(-1 - 4x_i(1 - 2x_i^2 - t_i))$$

Example 2

$$f(t, x) = 1 - 2x^2 - t, \quad t_0 = 0, \quad x_0 = 1, \quad h = 0.01$$

$$x_{i+1} = x_i + h(1 - 2x_i^2 - t_i) + \frac{h^2}{2}(-1 - 4x_i(1 - 2x_i^2 - t_i))$$

Step1:

$$x_1 = 1 + 0.01(1 - 2(1)^2 - 0) + \frac{(0.01)^2}{2}(-1 - 4(1)(1 - 2 - 0)) = 0.9901$$

Step2:

$$x_2 = 0.9901 + 0.01(1 - 2(0.9901)^2 - 0.01) + \frac{(0.01)^2}{2}(-1 - 4(0.9901)(1 - 2(0.9901)^2 - 0.01)) = 0.9807$$

Step3:

$$x_3 = 0.9716$$

Example 1

$$f(t, x) = 1 - 2x^2 - t, \quad t_0 = 0, \quad x_0 = 1, \quad h = 0.01$$

Summary of the results:

i	t_i	x_i
0	0.00	1
1	0.01	0.9901
2	0.02	0.9807
3	0.03	0.9716

Programming Euler Method

Write a MATLAB program to implement Euler method to solve:

$$\frac{dv}{dt} = 1 - 2v^2 - t. \quad v(0) = 1$$

$$\text{for } t_i = 0.01i, \quad i = 1, 2, \dots, 100$$

Programming Euler Method

```
f=inline('1-2*v^2-t','t','v')
h=0.01
t=0
v=1
T(1)=t;
V(1)=v;
for i=1:100
    v=v+h*f(t,v)
    t=t+h;
    T(i+1)=t;
    V(i+1)=v;
end
```

Programming Euler Method

`f=inline('1-2*v^2-t','t','v')` → Definition of the ODE

`h=0.01`

`t=0`

`v=1`

`T(1)=t;`

`V(1)=v;`

`for i=1:100`

`v=v+h*f(t,v)`

`t=t+h;`

`T(i+1)=t;`

`V(i+1)=v;`

`end`

Initial condition

Main Loop

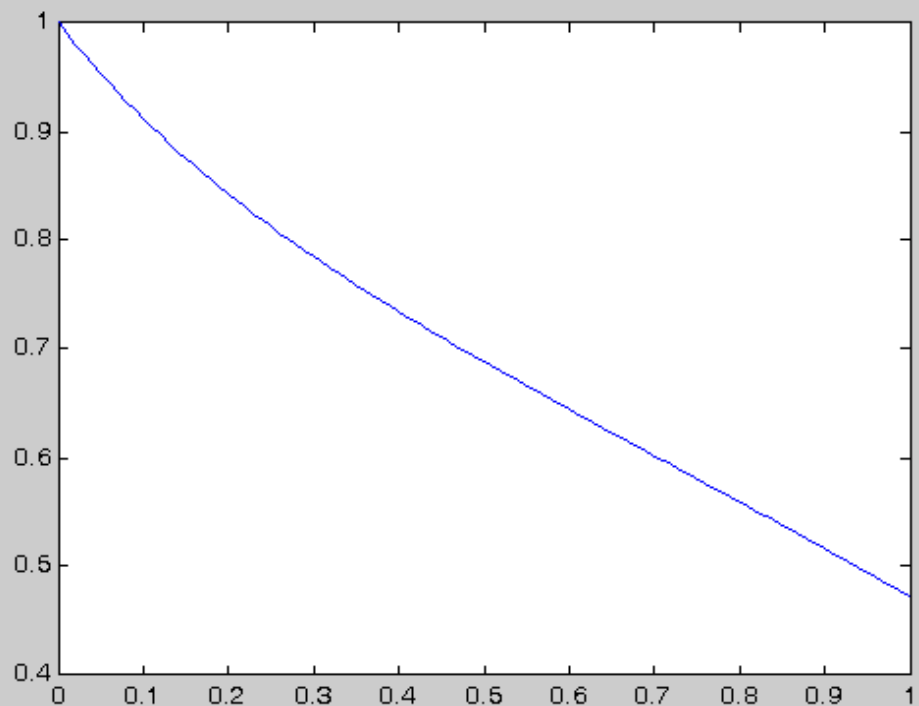
Euler method

Storing information

Programming Euler Method

Plot of the
solution

`plot(T,V)`



Next chapter

Midpoint method

Provide the accuracy of the second order Taylor series method without the need to calculate second order derivative.

Runge-Kutta methods

Provide the accuracy of high order Taylor series method without the need to calculate high order derivative.